

CBCS SCHEME

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BCS501

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Explain the domains of software applications.		08	L2	CO1
	b.	Outline the unique nature of WebApps.		08	L2	CO1
	c.	Explain various software myths. Discuss.		04	L2	CO1
OR						
Q.2	a.	Explain the activities performed in a software process framework?		06	L2	CO1
	b.	Explain the waterfall model along with its pros and cons.		08	L2	CO1
	c.	Explain specialized process models.		06	L2	CO1
Module – 2						
Q.3	a.	Explain how groundwork parameters are established in requirements engineering.		08	L2	CO2
	b.	What is the importance of quality function deployment in eliciting requirements?		06	L1	CO2
	c.	How can we validate requirements?		06	L1	CO2
OR						
Q.4	a.	Explain about scenario based modelling.		10	L2	CO2
	b.	Illustrate regarding how can we create a Behavioral Model.		10	L2	CO2
Module – 3						
Q.5	a.	Explain Agility along with the principles of Agility.		10	L2	CO3
	b.	Explain the Extreme Programming Process.		06	L2	CO3
	c.	Explain about the critics of XP.		04	L2	CO3
OR						
Q.6	a.	Explain the scrum flow process.		08	L2	CO3
	b.	Explain the communication principles guiding framework activity.		08	L2	CO3
	c.	How can we validate and test principles in coding.		04	L1	CO3
Module – 4						
Q.7	a.	Define Project. Show the contrast of software projects with other types of projects.		06	L2	CO4
	b.	Explain the ISO 12207 software development life cycle with a neat diagram.		10	L2	CO4
	c.	What are outsourced projects?		04	L1	CO4
OR						
Q.8	a.	Illustrate the cost benefit evaluation techniques.		10	L2	CO4
	b.	Illustrate the concept of Risk evaluation.		10	L2	CO4
Module – 5						
Q.9	a.	Explain the details to be drafted for achieving quality in software.		06	L2	CO5
	b.	Explain the software quality characteristics of ISO 9126.		08	L2	CO5
	c.	Explain process requirements for the process quality management.		06	L2	CO5
OR						
Q.10	a.	Explain about the decomposition techniques.		10	L2	CO5
	b.	Explain the COCOMO II model.		10	L2	CO5

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1		M	L	C
a.	Explain software process and software engineering practices.	10	L2	CO1
b.	Explain the waterfall model and incremental model, with diagram.	10	L2	CO1
OR				
a.	Explain Boehm Spiral process model with a neat diagram. Mention its advantages and disadvantages.	10	L2	CO1
b.	Explain the five activities of a generic process framework for software engineering.	10	L2	CO1
Module – 2				
a.	Explain the distinct tasks of requirement engineering.	10	L2	CO2
b.	Illustrate the UML use case diagram for safe home system.	10	L2	CO2
OR				
a.	Explain Class-Responsibility-Collaborator(CRC) modeling and data modeling with an example.	10	L2	CO2
b.	Explain the elements of analysis model in requirement modeling.	10	L2	CO2
Module – 3				
a.	Explain the principles of agile process development.	10	L2	CO3
b.	Explain the following : i) Adaptive software development ii) SCRUM	10	L2	CO3
OR				
a.	Explain the concepts of extremes programming with a neat diagram.	10	L2	CO3
b.	Explain design modeling principles that guide the respective framework activity.	10	L2	CO3
Module – 4				
a.	Illustrate the project management life cycle with a neat diagram.	10	L2	CO4
b.	Explain : i) Different ways of categorizing software projects ii) Smart objectives	10	L2	CO4
OR				
a.	Explain the difference between traditional versus modern project management practices along with the role of management.	10	L3	CO4
b.	Explain software development life cycle (ISO 12207) with a neat diagram.	10	L2	CO4
Module – 5				
a.	Explain Quality Management System with principles of BS EN ISO-9001-2000.	10	L2	CO5
b.	Explain the following : i) McCall model ii) Garvin's Quality Dimensions.	10	L2	CO5
OR				
a.	Describe six generic functions allowed in automated estimation techniques of software projects.	10	L3	CO5
b.	Explain COCOMO II model.	10	L2	CO5



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BCS501

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025

Software Engineering & Project Management

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain the software process in software engineering highlighting the importance of software engineering.	10	L2	CO1
	b.	Explain the five activities that a generic process framework for software engineering encompasses.	10	L2	CO1
OR					
Q.2	a.	Explain software myths with examples.	10	L2	CO1
	b.	Explain Incremental process models and evolutionary process models with a neat diagram.	10	L2	CO1
Module – 2					
Q.3	a.	Explain the different tasks which requirements engineering encompasses.	10	L2	CO2
	b.	Explain the nature and characteristics of software system.	10	L2	CO2
OR					
Q.4	a.	Explain requirements elicitation and various techniques used in requirements elicitation along with its importance.	10	L2	CO2
	b.	Illustrate an UML use case diagram for home security function.	10	L2	CO2
Module – 3					
Q.5	a.	Explain Agile process and agility principles.	10	L2	CO3
	b.	Explain Extreme Programming (XP) with a neat diagram.	10	L2	CO3
OR					
Q.6	a.	Explain SCRUM process with a neat diagram.	10	L2	CO3
	b.	Explain Agility with the cost of change with diagram. Explain the principles of Agile software development.	10	L2	CO3
Module – 4					
Q.7	a.	Explain different categories of software projects with example.	10	L2	CO4
	b.	Compare between Project Management Life Cycle And Software Development Life Cycle and its phases.	10	L2	CO4
OR					
Q.8	a.	Explain the difference between traditional and modern project management.	10	L2	CO4
	b.	Explain the concepts in activity planning in software project management.	10	L2	CO4
Module – 5					
Q.9	a.	Explain place of software quality in project management.	10	L2	CO5
	b.	Explain in detail the techniques to enhance software quality.	10	L1	CO5
OR					
Q.10	a.	Explain Quality Management Systems. With principles of BSENISO9001 : 2000.	10	L2	CO5
	b.	Explain the techniques to enhance software quality and software reliability. Explain SEICMM levels.	10	L2	CO5

Model Question Paper-1 with effect from 2022(CBCS Scheme)

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Fifth Semester B.E. Degree Examination Software Engineering and Project Management

TIME: 03 Hours

Max.Marks:100

Note:1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module-1			*Bloom's Taxonomy Level	COs	Marks
Q.01	a	Define process and explain generic process framework for software engineering.	L1	CO1	10
	b	Explain the different attributes are encountered in the vast majority of WebApps.	L2	CO1	10
OR					
Q.02	a	Compare and contrast Waterfall model and spiral model	L1	CO1	10
	b	Discuss the David Hooker's seven principles of software engineering practice.	L2	CO1	10
Module-2					
Q.03	a	With an example, describe the Class-Responsibility-Collaborator(CRC) modeling.	L1	CO2	10
	b	Explain three types of QFD with examples.	L2	CO2	10
OR					
Q.04	a	Explain scenario based model with example.	L2	CO2	10
	b	How can you develop an effective use case? Develop a UML use case diagram for home security function.	L1	CO2	10
Module-3					
Q.05	a	Elucidate the concepts of extreme programming(XP) with its functional diagram	L2	CO3	10
	b	Elucidate SCRUM process with a neat diagram	L1	CO3	10
OR					
Q.06	a	What is Agility? Explain Agility with the cost of change with Diagram. Explain the Principles of Agile Software Development	L1	CO3	10
	b	Explain Safe Home Security functions for Flow Oriented Modeling and Behavioral Modeling.	L1	CO3	10
Module-4					
Q.07	a	Explain the procedure of setting objectives for successful completion of software project.	L2	CO4	10
	b	With example explain different categories of Software Projects.	L2	CO4	10
OR					
Q.08	a	Elucidate the concepts in activity planning in software project management.	L1	CO4	10
	b	Explain the Risk Evaluation with example.	L3	CO4	10
Module-5					
Q.09	a	Define software quality and explain place of software quality in project management.	L2	CO5	10
	b	Explain capability process model and CMM key areas.	L1	CO5	10

OR					
Q.10	a	Explain Quality Management Systems with Principles of BSENISO 9001:2000	L1	CO5	10
	b	List and Explain the Techniques to enhance Software Quality and Software Reliability.	L3	CO5	10

Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.